Batterjee Medical College
For Science and Technology

Travel Medicine (Pt. 2)

An approach to a Case of Fever

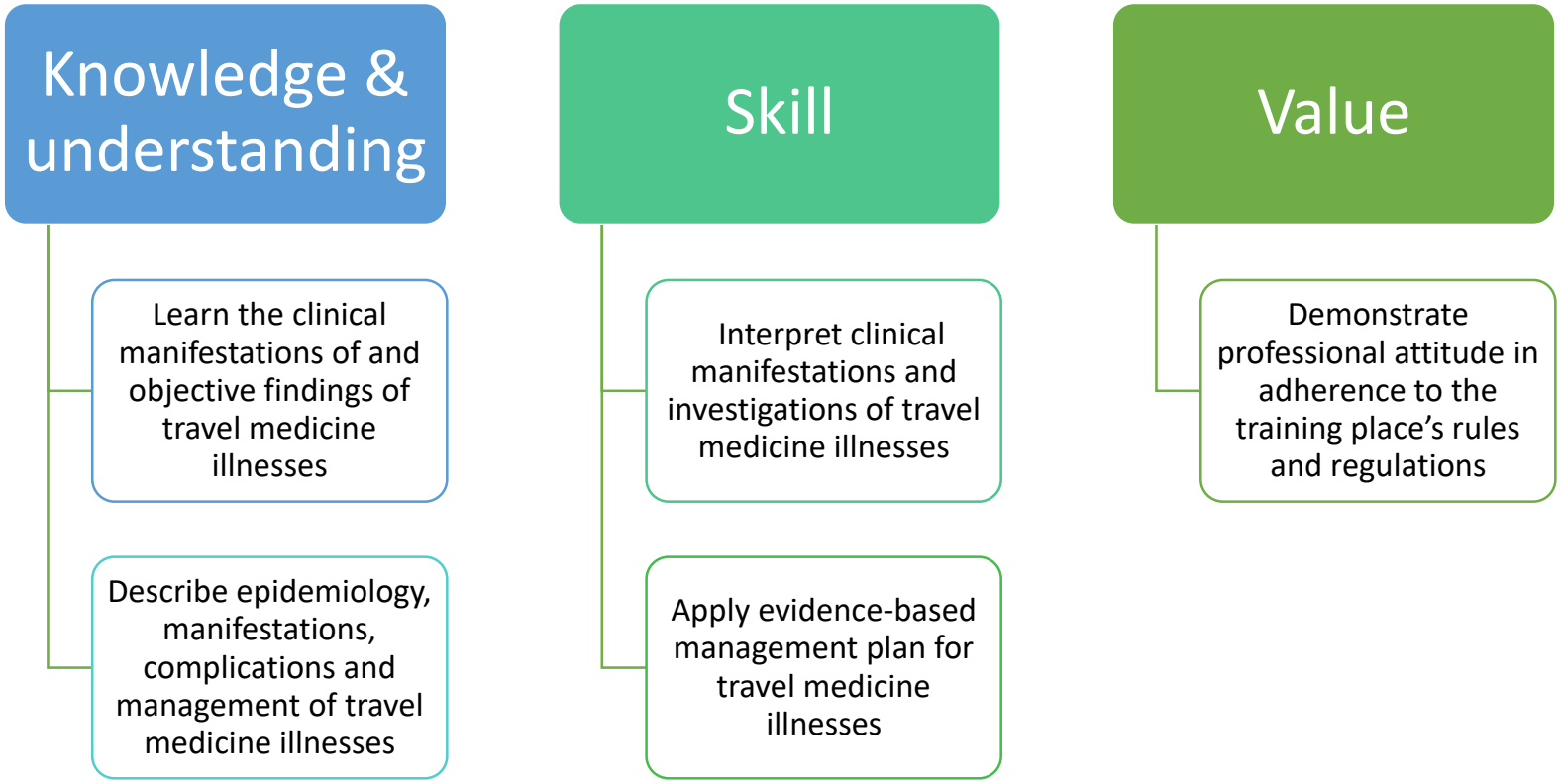
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BATTERJEE MEDICAL COLLEGE

BMC
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- Dengue Fever
- Zika Virus
- Chikungunya Fever
- Brucellosis

Learning Objectives



The Case

A 26-year-old primigravid woman at 22 weeks' gestation comes to the emergency department because of vaginal bleeding and suprapubic pain for the last 2 hours. She has no history of medical illness. She takes folic acid and a multivitamin supplement. The patient works as a physician in a local clinic. She has no history of recent travel. Her husband works as a flight attendant for an international airline and frequently travels to South America and Southeast Asia. Ultrasonography shows absent fetal cardiac activity and detachment of the placenta from the uterus. Examination of the stillborn fetus shows microcephaly. A reverse transcriptase polymerase chain reaction test of placental tissue confirms infection with a flavivirus. Which of the following measures during pregnancy is most likely to have prevented this fetus's condition?

a. Avoiding consumption of unpasteurized dairy products
b. Use of condoms
c. Administration of combined antiretroviral therapy
d. Avoiding consumption of undercooked meat
e. Administration of vaccine

Keep this case in mind.

We'll build the framework, then come back and solve it together at the end.

What caused the fetal complication & Which of the following measures during pregnancy is most likely to have prevented this fetus's condition?

Dengue Fever - Epidemiology

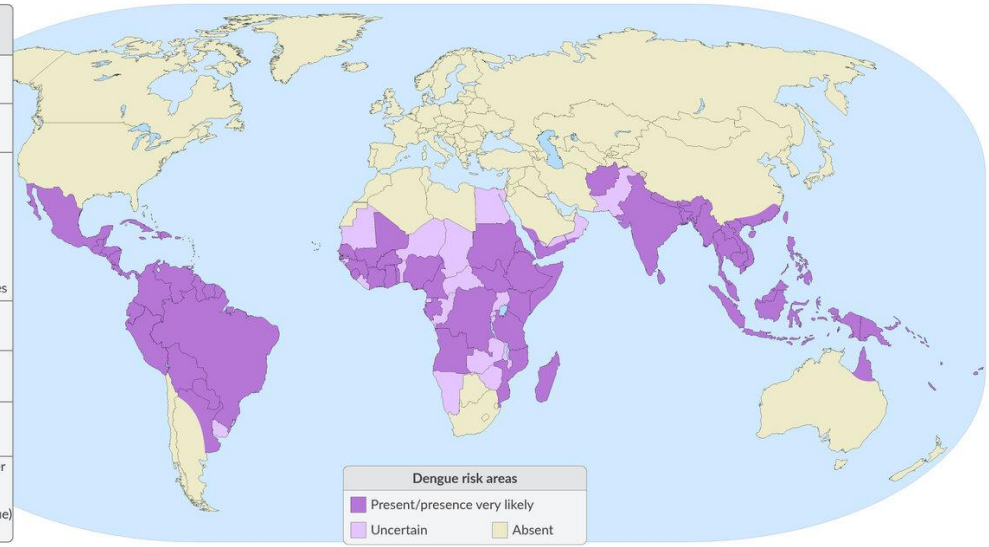
WHERE

Tropical and subtropical regions, including travel-related and locally acquired disease in suitable settings.

WHY IT MATTERS

Exposure geography and timing help interpret acute febrile illness after travel.

Dengue virus infections *Dengue virus (DENV) 1–5*	
Vector	- Mosquitoes (mostly Aedes aegypti)
Incubation period	- 2 to 14 days
Clinical features	- 90% asymptomatic **Classic dengue fever** : 1. Flu-like symptoms, joint and muscle pain, "break-bone fever", headache 2. Second fever peak, exanthema 3. Resolution of symptoms in most cases **Dengue hemorrhagic fever (1-2%)** : Bleeding, petechiae, pleural effusion, ascites
Annual incidence	- 50 to 100 million symptomatic cases
Mortality rate	- Over 15% in severe course, otherwise <1%
Prevention	- Vector control - Vaccination
Note	- Severe course with frequent hemorrhagic fever and recurrent illness - Dengue virus infections (dengue, dengue-like illness, and severe dengue) are nationally notifiable conditions

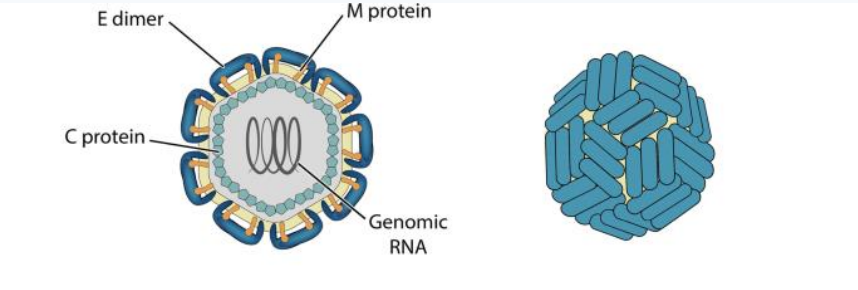


Dengue Fever - Etiology

PATHOGEN

Dengue virus (DENV 1–4)

RNA virus · genus Flavivirus



VECTOR

Aedes mosquitoes, particularly Aedes aegypti



Dengue Fever - Classification

01

WITHOUT WARNING SIGNS

Fever + suggestive symptoms

02

WITH WARNING SIGNS

Features of impending severe disease

03

SEVERE DENGUE

Plasma leakage, bleeding or organ injury

2009 WHO classification scheme[1]

Dengue without warning signs (D-W)

Fever and two of the following:
- Nausea, vomiting
- Rash
- Aches and pains
- Leukopenia
- Positive tourniquet test
- Laboratory confirmed dengue

With ability:
- To tolerate adequate volumes of oral fluid replacement
- To pass urine at least once every 6 hours

Dengue with warning signs (D+W)

Patients with any of the following features:
- At least one of the following warning signs:
  - Abdominal pain or tenderness
  - Persistent vomiting
  - Clinical fluid accumulation
  - Mucosal bleed (gingival bleeding, epistaxis, conjunctival bleeding, hematemesis, melena, fresh blood per rectum, hematuria, or vaginal bleeding)
  - Lethargy/restlessness
  - Liver enlargement >2 cm
  - Increased HTC with concurrent decrease in platelet count ($\leq$100,000 platelets/mm^3).

OR
- At least one comorbid condition such as:
  - Pregnancy
  - Infancy
  - Old age
  - Diabetes mellitus
  - Renal failure

OR
-Social circumstances such as
  - Living alone
  - Living far from hospital

Severe dengue fever (SDF)

Patients with any of the following features:
-Severe plasma leakage leading to:
  - Shock
  - Fluid accumulation leading to respiratory distress

-Severe bleeding as evaluated by clinician
-Severe organ involvement:
  - Liver: AsT or AlT > = 1000 IU/L
  - CNS: impaired consciousness.
  - Heart and other organs.

Dengue - Clinical Features

FEBRILE

Days 1–3 (variable)

High fever, pain, rash

CRITICAL

Around defervescence

Leakage, shock, rising Hct

RECOVERY

Fluid reabsorption

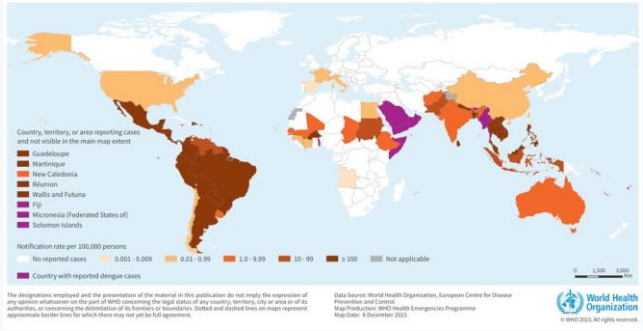
Clinical stabilization

Defervescence does NOT automatically mean improvement.

Clinical Features – Without Warning Signs

CONTEXT

Fever + residence in, or travel to, an endemic area



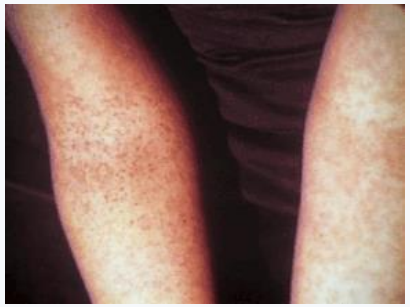
TYPICAL FEATURES

Nausea / vomiting

Rash

Headache or retro-orbital pain

Myalgia / arthralgia



OTHER FINDINGS

Leukopenia

Positive tourniquet test



Suspect dengue when fever is accompanied by ≥2 compatible findings.

Clinical Features – With Warning Signs

ABDOMINAL	Pain or tenderness
LEAKAGE	Ascites or pleural effusion
MENTAL STATE	Lethargy or restlessness
CBC TREND	Rising hematocrit + falling platelets
VOMITING	Persistent vomiting
BLEEDING	Mucosal bleeding
HEPATOMEGALY	Liver enlargement >2 cm

Warning signs often emerge around defervescence. Reassess even when fever improves.

Clinical Features – Severe Dengue

PLASMA LEAKAGE

Shock or respiratory distress from significant fluid accumulation

SEVERE BLEEDING

Clinically significant hemorrhage

ORGAN INVOLVEMENT

Severe hepatic, neurologic, cardiac or other organ dysfunction

Severe dengue is more likely after heterologous secondary infection, but can occur with primary infection.

Dengue Fever - Diagnostics

ROUTINE LABS

Leukopenia / neutropenia

Thrombocytopenia

Elevated AST

Hematocrit trend (plasma leakage)

CONFIRMATION

First 0–7 days: NAAT (RT-PCR) or NS1 antigen, with IgM

After day 7: IgM is often the primary test

Single IgG is NOT useful to diagnose acute dengue

Dengue Fever - Management

NO WARNING SIGNS

Oral hydration

Acetaminophen

Close follow-up and return precautions

WARNING / SEVERE

Observe or admit as appropriate

Carefully titrated IV isotonic fluids

Monitor vital signs, urine output, Hct

Blood products for significant bleeding

Avoid aspirin / NSAIDs. Do not give prophylactic platelet transfusions.

Zika Virus - Epidemiology

GEOGRAPHY

Tropical and subtropical areas

Historical outbreaks in Africa, Southeast Asia, Pacific islands and the Americas



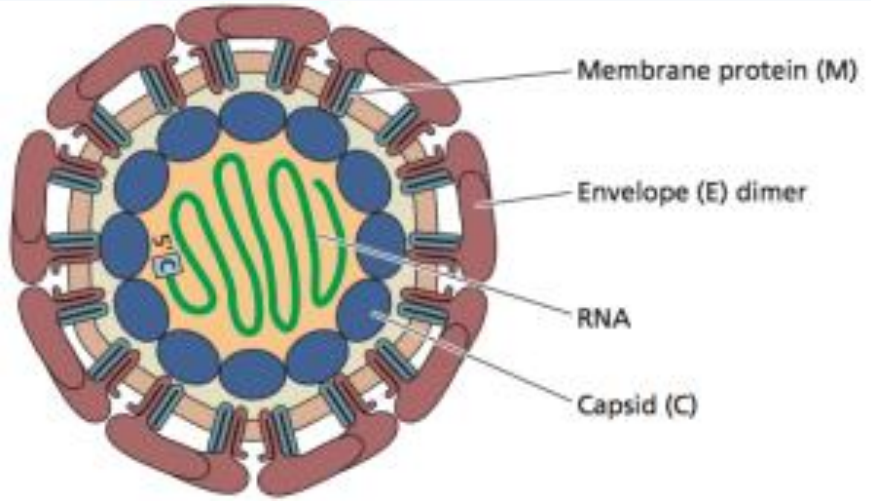
HISTORICAL CONTEXT

Large outbreaks in the Americas began in 2015, notably Brazil. Current risk must be assessed against up-to-date travel advisories.

Zika Virus - Etiology

PATHOGEN

- Zika virus
- Genus: flavivirus, type of arbovirus. Positive-sense, single-stranded, enveloped RNA



Zika Virus – Transmission

01

MOSQUITO

Aedes aegypti

02

PREGNANCY

Transplacental → fetus

03

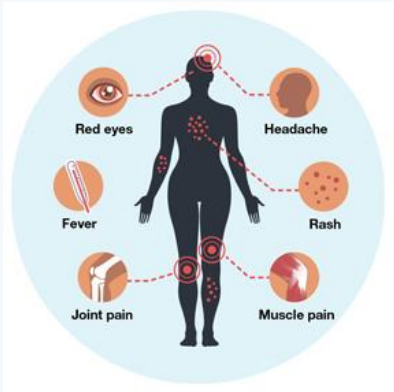
SEXUAL

Partner-to-partner

Zika Virus - Clinical Features

MANY ASYMPTOMATIC

A large proportion of infections cause no recognized symptoms.



SYMPTOMS

Low-grade fever

Headache, myalgia, arthralgia

Nonpurulent conjunctivitis

Malaise

RASH

Often maculopapular and pruritic

Zika Virus - Complications

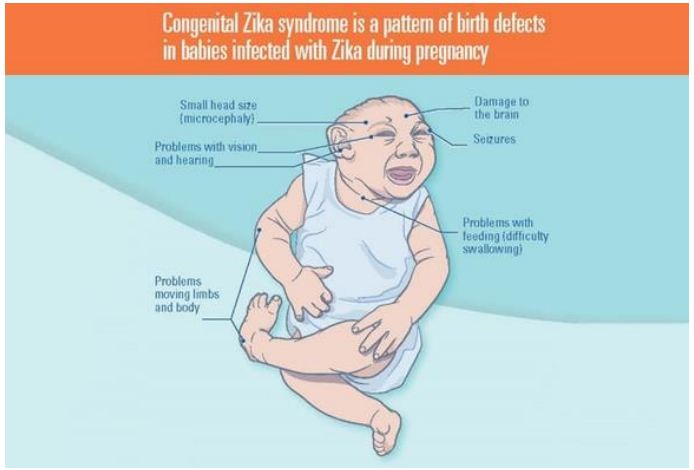
PREGNANCY

Congenital Zika syndrome

Microcephaly / CNS abnormalities

Growth restriction

Pregnancy loss



NEUROLOGIC

Guillain–Barré syndrome in infected adults

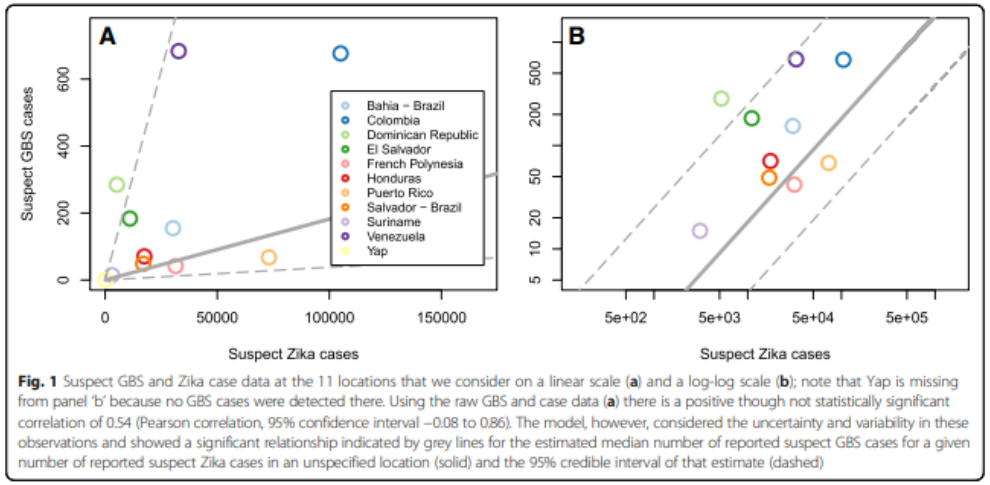


Fig. 1 Suspect GBS and Zika case data at the 11 locations that we consider on a linear scale (**a**) and a log-log scale (**b**); note that Yap is missing from panel 'b' because no GBS cases were detected there. Using the raw GBS and case data (**a**) there is a positive though not statistically significant correlation of 0.54 (Pearson correlation, 95% confidence interval −0.08 to 0.86). The model, however, considered the uncertainty and variability in these observations and showed a significant relationship indicated by grey lines for the estimated median number of reported suspect GBS cases for a given number of reported suspect Zika cases in an unspecified location (solid) and the 95% credible interval of that estimate (dashed)

Mier-Y-Teran-Romero L, Delorey MJ, Sejvar JJ, Johansson MA. Guillain-Barré syndrome risk among individuals infected with Zika virus: a multi-country assessment. BMC Med. 2018 May 15;16(1):67. doi: 10.1186/s12916-018-1052-4. PMID: 29759069; PMCID: PMC5952697.

Zika Virus - Management

TREATMENT

No specific antiviral therapy

Supportive care

Assess for concomitant dengue before using NSAIDs

PREGNANCY & TRAVEL

Avoid travel to areas with active Zika notices during pregnancy

Use condoms or abstain after possible exposure

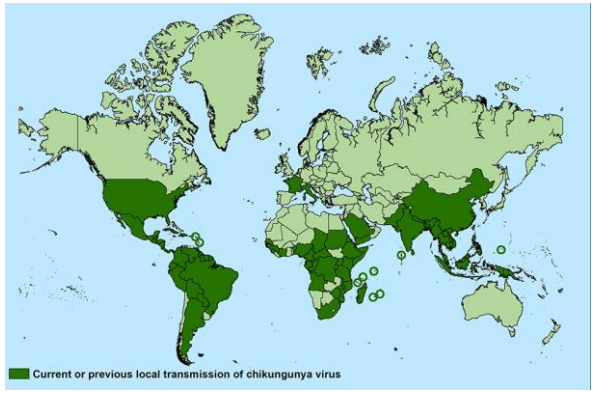
Provide preconception counseling based on current guidance

Consider molecular / serologic tests based on symptoms, pregnancy status and exposure.

Chikungunya Fever - Overview

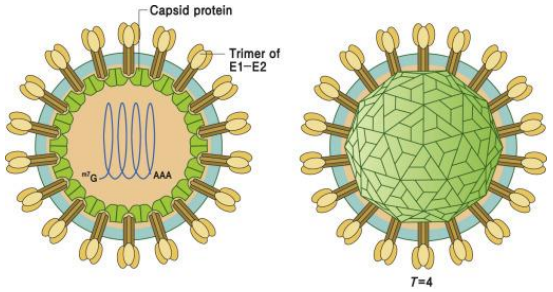
EPIDEMIOLOGY

Tropical / subtropical distribution with sporadic outbreaks



VIROLOGY

Chikungunya virus · family Togaviridae · alphavirus



EXPOSURE

Aedes aegypti and Aedes albopictus

Dengue coinfection can occur





Chikungunya Fever – Clinical Features & Diagnostics

CLASSIC CLUES

Acute fever / malaise

Severe polyarthralgia

Headache, myalgia

Possible lymphadenopathy

DIAGNOSIS

Exposure + clinical presentation

RT-PCR during early illness

Serology later in the illness

Joint symptoms may persist for months or longer despite resolution of fever.

Chikungunya Fever – Management

ACUTE MANAGEMENT

Supportive care

Hydration, rest, analgesia

Exclude dengue before using NSAIDs

COMPLICATION

Persistent arthralgia or inflammatory arthritis may require follow-up

Most acute infections improve, but chronic musculoskeletal symptoms are a key distinction.

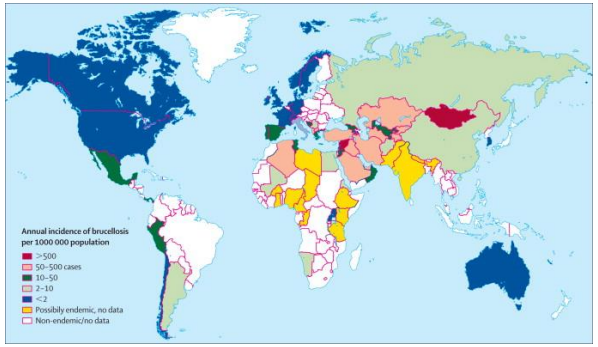
Brucellosis – Overview

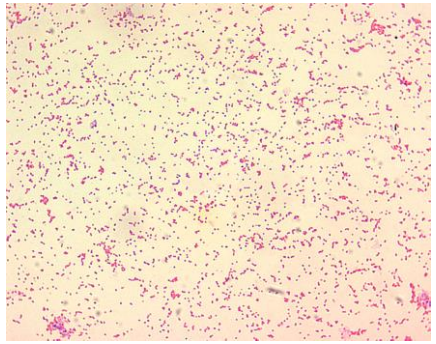
BACTERIUM

Brucella species

Small gram-negative coccobacilli

Zoonotic infection





EXPOSURES

Unpasteurized dairy products

Contact with infected animals / animal tissues

Occupational exposure (e.g., veterinarians)



In Saudi clinical practice, ask specifically about raw milk and livestock exposure.

Brucellosis – Clinical Features & Diagnosis

SYSTEMIC

Flu-like illness

Night sweats

High or undulant fever

Lymphadenopathy



FOCAL DISEASE

Arthralgia / arthritis

Spondylitis / osteomyelitis

Less often neurobrucellosis or endocarditis



A focal spinal or cardiac presentation should prompt assessment for complications.

Brucellosis – Management

DIAGNOSIS

Serologic testing: screening and confirmatory assays (including agglutination)

Blood cultures when indicated

Inform the laboratory of suspected Brucella!

TREATMENT

Uncomplicated adult disease: doxycycline + rifampin, usually ≥6 weeks

Regimens vary with pregnancy, age, and focal complications

Approach to Fever

Key History:

- Severity
- Duration
- Pattern
- associated localizing symptoms such as rash, wheezing, cough, and ear discharge, etc.
- Night sweats, weight loss
- Travel history
- Contact with sick persons
- Contact with animals
- Immunizations
- Recent surgery
- Medications

Approach to Fever

Key Physical:

- Vital signs
- HEENT
- Neck
- Heart
- Lung
- Abdominal
- Skin

Approach to Fever

DDx:

Infections
Tuberculosis (especially extrapulmonary)
Abdominal abscesses
Pelvic abscesses
Dental abscesses
Endocarditis
Osteomyelitis
Sinusitis
Cytomegalovirus
Epstein-Barr virus
Human immunodeficiency virus
Lyme disease
Prostatitis
Sinusitis
Malignancies
Chronic leukemia
Lymphoma
Metastatic cancers
Renal cell carcinoma
Colon carcinoma
Hepatoma
Myelodysplastic syndromes
Pancreatic carcinoma
Sarcomas
Sarcoidosis

Autoimmune conditions
Adult Still's disease
Polymyalgia rheumatica
Temporal arteritis
Rheumatoid arthritis
Rheumatoid fever
Inflammatory bowel disease
Reiter's syndrome
Systemic lupus erythematosus
Vasculitides
Miscellaneous
Drug-induced fever
Complications from cirrhosis
Factitious fever
Hepatitis (alcoholic, granulomatous, or lupoid)
Deep venous thrombosis

Approach to Fever

General Workup:

- CBC
- BMP
- LFTs
- UA
- Urine culutres
- CXR
- Sputum Cultures
- Stool analysis
- Viral antibodies

Specific Workup:

- ANA
- RF
- LP
- Serology
- Abdominal Imaging
- TB

To Summarize

DENGUE

KEY CLUE

Bleeding / capillary leak

TESTING

NS1 / NAAT early; IgM later

ZIKA

KEY CLUE

Conjunctivitis + pregnancy risk

TESTING

NAAT / serology per exposure

CHIKUNGUNYA

KEY CLUE

Severe polyarthralgia

TESTING

PCR early; serology later

BRUCELLOSIS

KEY CLUE

Raw dairy + night sweats

TESTING

Serology ± cultures

Case Scenarios



Case Scenario 1

A 26-year-old primigravid woman at 22 weeks' gestation comes to the emergency department because of vaginal bleeding and suprapubic pain for the last 2 hours. She has no history of medical illness. She takes folic acid and a multivitamin supplement. The patient works as a physician in a local clinic. She has no history of recent travel. Her husband works as a flight attendant for an international airline and frequently travels to South America and Southeast Asia. Ultrasonography shows absent fetal cardiac activity and detachment of the placenta from the uterus. Examination of the stillborn fetus shows microcephaly. A reverse transcriptase polymerase chain reaction test of placental tissue confirms infection with a flavivirus. Which of the following measures during pregnancy is most likely to have prevented this fetus's condition?

a. Avoiding consumption of unpasteurized dairy products
b. Use of condoms
c. Administration of combined antiretroviral therapy
d. Avoiding consumption of undercooked meat
e. Administration of vaccine

Case Scenario 2

A previously healthy 25-year-old man comes to the physician because of a 4-day history of fever, joint and body pain, diffuse headache, and pain behind the eyes. This morning he noticed that his gums bled when he brushed his teeth. He returned from a backpacking trip to the Philippines 4 days ago. His temperature is 39.4°C (103.0°F). Physical examination shows a diffuse maculopapular rash. His leukocyte count is 3,200/mm^3 and platelet count is 89,000/mm^3. Further evaluation shows increased serum levels of a flavivirus. Which of the following is the most likely causal pathogen?

a. Chikungunya virus
b. Zika virus
c. Hepatitis C virus
d. Dengue virus

Questions?



References & further reading

- Harrison's Principles of Internal Medicine 20th edition
- WHO Dengue Guidelines for Diagnosis, Treatment, Prevention and Control
- Amboss
- 1st aid for the internal medicine boards, 4th edition

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Thank You